Homework 1: Probability & Homo-transforms

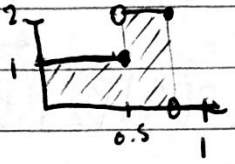
PR-Q1

a) $p(X=x) = \begin{cases} 0.2 & x=1 \\ 0.4 & x=2 \\ 0.5 & x=3 \\ 0 & \text{otherwise} \end{cases}$ not a valid PMF/PDF due to Axiom of Unitary/normalization

b) $p(X=x) = \begin{cases} 0.3 & x=1 \\ 0.4 & x=2 \\ 0.3 & x=3 \\ 0 & \text{otherwise} \end{cases}$ PMF

c) $p(X=x) = \begin{cases} 0.8 & x=1 \\ 0.3 & x=2 \\ -0.1 & x>3 \\ 0 & \text{otherwise} \end{cases}$ not PMF/PDF due to Axiom of non-negativity

d) $p(X=x) = \begin{cases} 1 \text{ if } 0 \le x \le 0.5 \\ 2 - 2x \text{ if } 0.5 < x \le 0.75 \\ 0 \text{ otherwise} \end{cases}$



PDF

e) Any $f_x(x)$ where, given $a<b<c$

$$\int_a^c f_x(x) \neq \int_a^b f_x(x) + \int_b^c f_x(x)$$

not PMF/PDF due to Axiom of Additivity

f) $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ PDF

$\int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-\frac{x^2}{2}}$??

$\frac{1}{2\pi}$

4

PR-Q2

a) ~~P(A|B)=0.8 b) P(A)~~ $P(A \cup 6) = 0.3$

b) $P(C \cup 6hr) = 0.01$

c) marginalize out time

$P_g(A) = \sum_h P_{g,h}(A,h) = \boxed{0.44}$ $P_g(B) = 0.35$ $P_g(C) = 0.21$

d) $P(A|6) = \frac{P(A,6)}{P(6)} = \frac{0.3}{0.41} = \boxed{73.17\% \text{ or } .7317}$

PR-Q3

- Independance means the prob. of one random variable is not influenced by knowledge of value of another random variable
- Cond indep. occurs when knowledge of 3rd event means the other 2 events are still independant.

ex of indep ≠ cond indep



if C is impacted by just A & B & is known, you can determine A from B, but normally A & B are independant

PR-Q4

$P(x|y) = \frac{P(y|x)P(x)}{P(y)} = \frac{P(y|x)P(x)}{\sum_x P(y|x)P(x)}$

$P(A|B) = \frac{P(A,B)}{P(B)}$ → law of cond prob.

$P(A) = \sum_B P(A|B)P(B)$ → law of tot. prob

$P(x|y) = \frac{P(x,y)}{P(y)}$ → $P(y) = \sum_x P(y|x)P(x)$

$P(x|y) = \frac{P(x,y)}{\sum_x P(y|x)P(x)}$ → $P(y|x) = \frac{P(x,y)}{P(x)}$

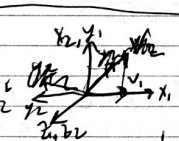
$P(x,y) = P(y|x)P(x)$

$P(x|y) = \frac{P(y|x)P(x)}{\sum_x P(y|x)P(x)}$

Rot-Mat's

RM-Q1

Show $V_1^a \cdot V_2^a = V_1^b \cdot V_2^b$



$V_1' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $V_1^2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$

$V_2' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $V_2^2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

also the dot product is just the amount $V_1' \cdot V_2' = 1$
the vectors point in the same direction $V_1^2 \cdot V_2^2 = 1$
(scalar value

RM-Q2

Show $\|v\| = \|Rv\|$

$V_1' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $V_1^2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = R_1^2 V_1'$

$\|V_1'\| = 1 = \|V_1^2\| = 1$

RM-Q3

Show $\|P_1 - P_2\| = \|RP_1 - RP_2\|$

$V_1' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $V_2' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$V_1^2 = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$ $V_2^2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

$R_1^2 V_1'$ $R_1^2 V_2'$

$V_1' - V_2' = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$ $V_1^2 - V_2^2 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$

$\|V_1' - V_2'\| = 1$ $\|V_1^2 - V_2^2\| = 1$ ✓

RM-Q4

$R^T R = I$

$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ rotate → $R_2^T(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$R_2(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = 1$

$= 1 \quad = 1 \quad = 1$

$r_1 \cdot r_2 = 0$ $c_1 \cdot c_2 = 0$

$r_1 \cdot r_3 = 0$ $c_1 \cdot c_3 = 0$

$r_2 \cdot r_3 = 0$ $c_2 \cdot c_3 = 0$

RM-Q5

a) $\det(R) = \pm 1$

b) $\det(R) = r_3 \cdot (r_1 \times r_2)$

$\det(R^T R) = \det(I) = 1$

$\det(R^T R) = \det(R^T)\det(R) = \det(R^2) = 1$

$\det(R) = \pm 1$ ✓✓

RM-Q6

$x_1 * x_2 \in X$ For all $x_1, x_2 \in X$

$(x * x_2) * x_3 = x_1 * (x_2 * x_3)$

$I \in X$ s.t. $I * x = x * I$ For all $x \in X$

$x \in X$ exists $y \in X$ s.t. $x * y = y * x = I$

$SO(n) = \{R \mid R^T R = I, \det R = 1\}$ w matrix multiplication

1) $R_1, R_2 \in SO(n)$: $(R_1 R_2)^T (R_1 R_2) = R_2^T R_1^T R_1 R_2 = I$ ✓
2) $(R_1 R_2) R_3 = R_1 (R_2 R_3)$ ✓
3) $IR = RI$ ✓✓
4) $R^{-1} = R^T$ ✓✓

RM-Q7

$A = \begin{bmatrix} c\theta & -s\theta \\ s\theta & c\theta \end{bmatrix}$ $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $A^T A = I$ $\det(A) = 1$

$a^2 + c^2 = 1$ $ab + cd = 0$

$b^2 + d^2 = 1$

$\sin^2\theta + \cos^2\theta = 1$

$a = \cos\theta$ $c = \sin\theta$

$b = \sin\theta$ $d = \cos\theta$

RM-Q8

a) $R_z(0) = I$ $\begin{bmatrix} c(0) & -s(0) & 0 \\ s(0) & c(0) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$ ✓

b) $R_{z\theta} R_{z\phi} = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$R_{z\,\theta+\phi} = \begin{bmatrix} c(\theta+\phi) & -s(\theta+\phi) & 0 \\ s(\theta+\phi) & c(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} c\theta c\phi - s\theta s\phi & -s\theta c\phi - s\phi c\theta & 0 \\ s\theta c\phi + s\phi c\theta & -s\theta s\phi + c\theta c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c(\theta+\phi) & -s(\theta+\phi) & 0 \\ s(\theta+\phi) & c(\theta+\phi) & 0 \\ 0 & 0 & 1 \end{bmatrix}$

trig identities

c) $(R_{z\theta})^{-1} = R_z(-\theta)$

$(R_{z\theta})^{-1} = (R_{z\theta})^T$

$R_{z\theta} = \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_{z\theta}^T = R_{z\theta}^{-1} = \begin{bmatrix} c\theta & s\theta & 0 \\ -s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$R_{z-\theta} = \begin{bmatrix} c\theta & s\theta & 0 \\ -s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

RM-Q9

~~$R_x R_y R_z$~~ $R_{z\theta} R_y R_{x\phi}$

* world-front — important
current-back

RM-Q10

$R_x R_{x\phi} R_{z\theta}$

4

Homo-Xforms

HT-Q1

Find T_3^0, T_2^0, T_1^0, T_2^3

$T_1^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $T_2^0 = \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & 1.5 \\ 0 & 0 & 1 & 1.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ assume coordinate frame in center of cube (2cm sides)

$T_3^0 = \begin{bmatrix} 0 & 1 & 0 & -0.5 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $T_2^3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1.9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

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HT-Q2

if frame 3 $R_{z_7}(90^\circ)$

$T_3^0 = \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & -1 & 0 & 1.5 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $T_2^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1.9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ ✓

HT-Q3

frame 2 $R_{z_2}(90^\circ)$ + $T: \begin{bmatrix} 0 \\ 0.8 \\ 0.1 \end{bmatrix}$ relitive

Find T_{2b}^3 + T_3^{2b}

~~T_{4b}^0~~ ~~T_{2b}^0~~ $T_{2b}^3 = T_2^3 T_{2b}^2$

$T_{2b}^1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ +1 & 0 & 0 & 0.8 \\ 0 & 0 & 1 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$T_{2b}^3 = T_1^0 T_{2b}^1 T_0^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0.8 \\ 0 & 0 & 1 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & -1.5 \\ 1 & 0 & 0 & 0.5 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} =$

$$T_{2b}^{3} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1.8 \\ 0 & 0 & 1 & 1.1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & -1.5 \\ 1 & 0 & 0 & 0.5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & -0.5 \\ 0 & 1 & 0 & 0.3 \\ 0 & 0 & -1 & 4.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_{2b}^{3} = \begin{bmatrix} 1 & 0 & 0 & 0.3 \\ 0 & -1 & 0 & 0.5 \\ 0 & 0 & -1 & 1.9 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad (T_{2b}^{3})^{-1} = T_{3}^{2b} = \begin{bmatrix} 1 & 0 & 0 & -0.3 \\ 0 & -1 & 0 & 0.5 \\ 0 & 0 & -1 & 1.9 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$-R^{T}d = -\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.5 \\ 1.9 \end{bmatrix} = \begin{bmatrix} -0.3 \\ 0.5 \\ 1.9 \end{bmatrix}$$